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Course: Compiler Design

Lab Task :3

Task 1: Create another C++ function `checkExpression2(string s)` in the same program that checks whether the input is **Valid** or **Invalid** for only these forms:

Valid forms: `X++`, `++X`, `X--`, `--X`

Invalid examples: `--X`, `+X` Assume `X` is a single letter variable.

Sample Input 1: `X++`

Expected Output 1: Valid

Sample Input 2: `---X`

Expected Output 2: Invalid

Sample Input 3: `+X` **Expected Output 3:** Invalid

Answer:

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

void checkExpression2(string s) {
    if (s.length() != 3) {
        cout << "Invalid" << endl;
        return;
    }
    if (isalpha(s[0])) {
        if ((s[1] == '+' && s[2] == '+') || (s[1] == '-' && s[2] == '-')) {
            cout << "Valid" << endl;
            return;
        }
    }
    else if (isalpha(s[2])) {
        if ((s[0] == '+' && s[1] == '+') || (s[0] == '-' && s[1] == '-')) {
            cout << "Valid" << endl;
            return;
        }
    }
    cout << "Invalid" << endl;
}

int main() {
    string userInput;
    cout << "Enter an expression to check: ";
    cin >> userInput;
    cout << "Expected Output: ";
    checkExpression2(userInput);

    return 0;
}
```

Output:

```
"E:\Downloads\Compiler Mid" x + v
Enter an expression to check: +X-
Expected Output: Invalid

Process returned 0 (0x0)   execution time : 11.904 s
Press any key to continue.
|
```

```
"E:\Downloads\Compiler Mid" x + v
Enter an expression to check: --X
Expected Output: Valid

Process returned 0 (0x0)   execution time : 5.731 s
Press any key to continue.
```

```
"E:\Downloads\Compiler Mid" x + v
Enter an expression to check: ++X
Expected Output: Valid

Process returned 0 (0x0)   execution time : 10.805 s
Press any key to continue.
```

Task 2: Write a C++ program that takes one line as input and prints **Valid** or **Invalid** for a simple arithmetic expression using:

- operands as single letters only, like A, b, X
- operators only from + - * /
- spaces may appear anywhere

Validity rules:

1. Expression cannot start or end with an operator
2. No two operators can be consecutive
3. No two operands can be consecutive without an operator

Sample Input 1: A + B*C

Expected Output 1: Valid

Sample Input 2: A + * B

Expected Output 2: Invalid

Sample Input 3: AB + C

Expected Output 3: Invalid

Answer:

```
#include <iostream>
#include <string>
#include <cctype>

using namespace std;

bool validateExpression(string str) {

    bool expectOperand = true;
    bool hasSeenChar = false;

    for (int i = 0; i < str.length(); i++) {
        char c = str[i];
        if (isspace(c)) continue;
        hasSeenChar = true;

        if (expectOperand) {

            if (isalpha(c)) {
                expectOperand = false;
            } else {
                return false;
            }
        } else {

            if (c == '+' || c == '-' || c == '*' || c == '/') {
                expectOperand = true;
            } else {
                return false;
            }
        }
    }
    return hasSeenChar && !expectOperand;
}

int main() {
    string input;
    getline(cin, input);

    if (validateExpression(input)) {
        cout << "Valid" << endl;
    } else {
        cout << "Invalid" << endl;
    }

    return 0;
}
```

Output:

```
"E:\Downloads\Compiler Mid  ×  +  ∨  
a+*jk  
Invalid  
  
Process returned 0 (0x0)   execution time : 9.430 s  
Press any key to continue.
```

```
"E:\Downloads\Compiler Mid  ×  +  ∨  
d*f-b  
Valid  
  
Process returned 0 (0x0)   execution time : 12.101 s  
Press any key to continue.
```